Algebraic Geometry – Exercise Sheet 4

Week 4: Generic points, dimension and scheme-theoretic fibres

One Monday exercise session, followed by a revision bank; Lectures 15–18

Difficulty. $\star$ routine; $\star\star$ synthesis; $\star\star\star$ challenge or extension.
Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

Monday afternoon — material available: Lectures 1–16

Exercise 1. Components and generic points $\star\star$ **CORE**

Let

$$I = (xy, xz) \subset \Bbbk[x, y, z], \qquad X = \operatorname{Spec}(\Bbbk[x, y, z]/I).$$

(i) Show that $I = (x) \cap (y, z)$.

(ii) Determine the irreducible components and their generic points.

(iii) Is X reduced, connected, irreducible, or integral?

Exercise 2. The function field of the cusp $\star\star$ **CORE**

Let

$$C = \operatorname{Spec}\big(\Bbbk[x, y]/(y^2 - x^3)\big).$$

Show that C is integral, compute its function field, and deduce its dimension.

Exercise 3. Chains and dimensions $\star$ **CORE**

(i) Exhibit a maximal chain of irreducible closed subsets in $\operatorname{Spec}(\mathbb{Z})$.

(ii) Give a chain of length n in $\mathbb{A}^n_{\Bbbk}$.

(iii) Determine the dimension of

$$\operatorname{Spec}\big(\Bbbk[x, y, z]/(xy, xz)\big)$$

from its components.

Exercise 4. Noether normalisation and dimension $\star\star$ **CORE**

Let

$$B = \Bbbk[x, y]/(y^2 - x^3).$$

(i) Show that B is finite over $\Bbbk[x]$.

(ii) Use this to recover $\dim(B) = 1$.

(iii) Compare with the computation from the function field.

After Lectures 17–18 — revision bank on fibres

Exercise 5. The generic affine line $\star$ **CORE**

Let

$$\pi : \mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[t, x]) \longrightarrow \mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}(\mathbb{C}[t])$$

be the projection. Compute its fibre over a closed point $(t-a)$ and its fibre over the generic point (0). Explain why the latter is $\operatorname{Spec}(\mathbb{C}(t)[x])$ and verify the dimension formula for the generic fibre.

Exercise 6. A fibre product larger than the set-theoretic one $\star\star$ **CHOICE**

Let $X = \operatorname{Spec}(\mathbb{C})$ and $S = \operatorname{Spec}(\mathbb{R})$. Compute $X \times_S X$ and compare its points with the set-theoretic fibre product of the underlying one-point spaces.

Exercise 7. The power map and its fibres $\star\star$ **CORE**

Assume that $\Bbbk$ is algebraically closed of characteristic 0. Let

$$f : \mathbb{A}^1_{\Bbbk} \longrightarrow \mathbb{A}^1_{\Bbbk}, \qquad t \longmapsto t^n.$$

Compute every scheme-theoretic fibre, determine when it is reduced, compute the generic fibre, and show that f is finite.

Exercise 8. A family whose special fibre becomes reducible ⋆⋆ CORE

Let

$$X = \operatorname{Spec}\big(\Bbbk[x,y,t]/(xy-t)\big) \longrightarrow \mathbb{A}^1_t.$$

Compute the fibres, show that all have dimension 1, and determine which are integral and which are reducible.

Exercise 9. A jumping fibre ⋆⋆ CORE

Consider

$$f : \mathbb{A}^2_{\Bbbk} \longrightarrow \mathbb{A}^2_{\Bbbk}, \qquad (x,y) \longmapsto (x, xy).$$

Compute the fibre over (a,b) and show that the generic fibre has dimension 0, while the fibre over $(0,0)$ has dimension 1.

Exercise 10. The generic fibre formula in examples ⋆⋆ CHOICE

Verify

$$\dim X_\eta = \dim X - \dim Y$$

for the map in the preceding exercise and for the projection $\mathbb{A}^{m+n}_{\Bbbk} \to \mathbb{A}^m_{\Bbbk}$.

Optional outreach exercises: Hilbert schemes of points

Exercise 11. $\operatorname{Hilb}^n(\mathbb{A}^1)$ ⋆⋆ CHOICE

Assume that $\Bbbk$ is algebraically closed. Show that every ideal $I \subset \Bbbk[x]$ with $\dim_{\Bbbk} \Bbbk[x]/I = n$ is generated by a unique monic polynomial of degree n. Deduce

$$\operatorname{Hilb}^n(\mathbb{A}^1_{\Bbbk}) \simeq \mathbb{A}^n_{\Bbbk}$$

and explain how factorisation records support and multiplicities.

Exercise 12. Commuting matrices and a cyclic vector ⋆⋆⋆ OPTIONAL

Let V be n-dimensional and let (X,Y,v) satisfy $[X,Y]=0$ and $\Bbbk[X,Y]v = V$. Show that

$$I = \{f \in \Bbbk[x,y] \mid f(X,Y)v = 0\}$$

is an ideal and $\Bbbk[x,y]/I \simeq V$. Determine the triple for $I = (y, x^2)$.